Vertex formula

1

a. Axis of symmetry

b. Example answer:

The x-coordinate of the vertex of the parabola occurs at $x = \frac{-b}{2a}$. The y-coordinate of the vertex is found by substituting $x = \frac{-b}{2a}$ into the parabola's equation and evaluating the function at this value of x.

2 $(1, -12)$

3 $y = 0$

4 $x = \frac{1}{4}$

Vertex form

5

a. Infinitely many

b. Infinitely many

c. 1

d. $y = -\frac{3x^2}{25} - \frac{6x}{25} + \frac{72}{25}$

6

a. Vertex

b. Two

7 $y = (x+5)^2 - 48$

8

a. $y = (x+2)^2 - 4$

b. $(-2, -4)$

9

a. $(7, 8)$

b. $y = 8 + 3(x+7)^2$

c. $(-7, 8)$

10

a. $(3, 5)$

b. $y = (x-3)^2 + 5$

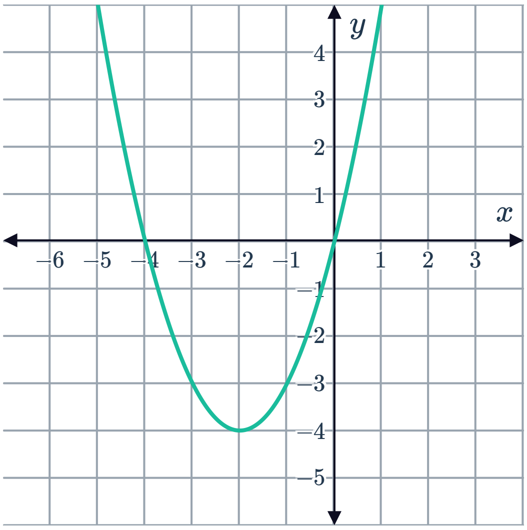
Graphs of quadratics

11

a

i $x = 0, -4$ **ii** $y = 0$ **iii** $x = -2$ **iv** $(-2, -4)$

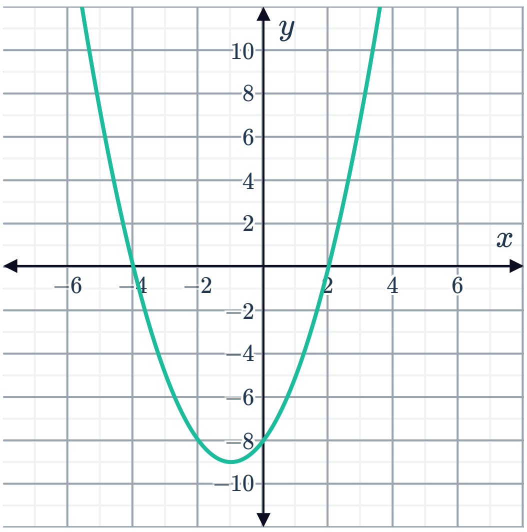
v



b

i $x = -4, 2$ **ii** $y - 8$ **iii** $x = -1$ **iv** $(-1, -9)$

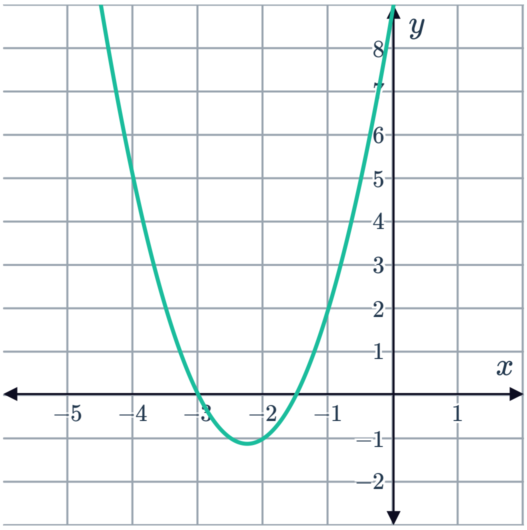
v



c

i $x = -\frac{3}{2}, -3$ **ii** $y = 9$ **iii** $x = -\frac{9}{4}$ **iv** $\left(0, \frac{9}{8}\right)$

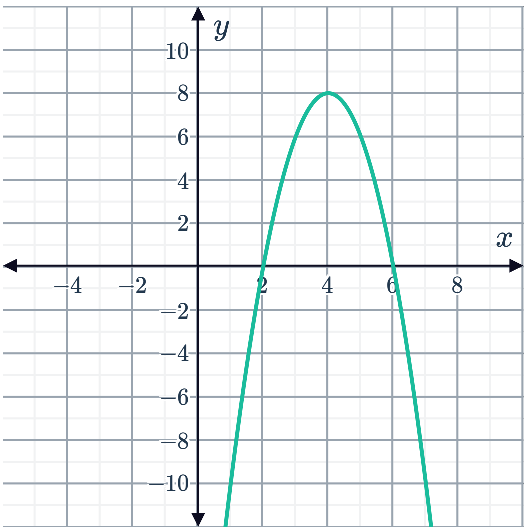
v



d

i $x = 2, 6$ **ii** $y = -24$ **iii** $x = 4$ **iv** $(4, 8)$

v



12

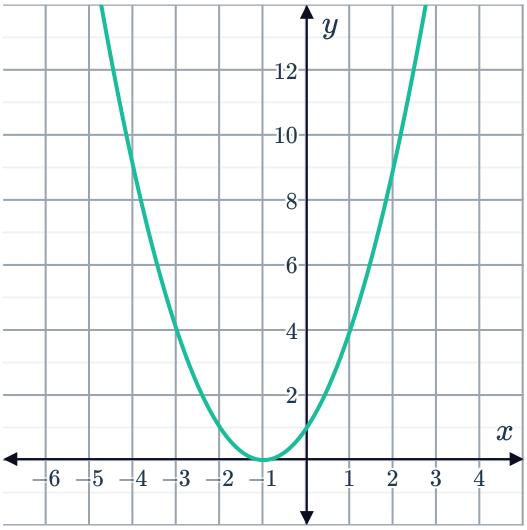
a

i $x = -1$

ii $y = 1$

iii $(-1, 0)$

iv



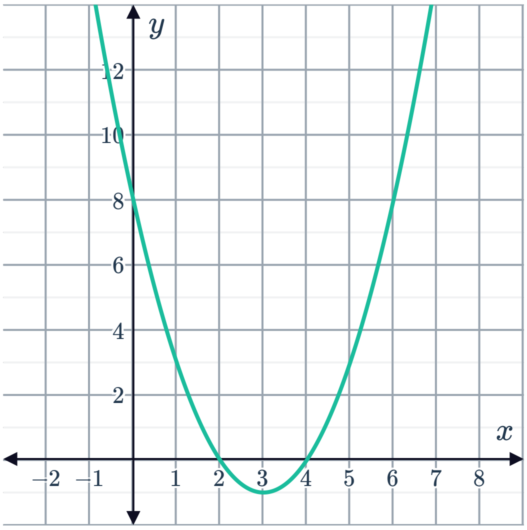
b

i $x = 4, 2$

ii 8

iii $(3, -1)$

iv



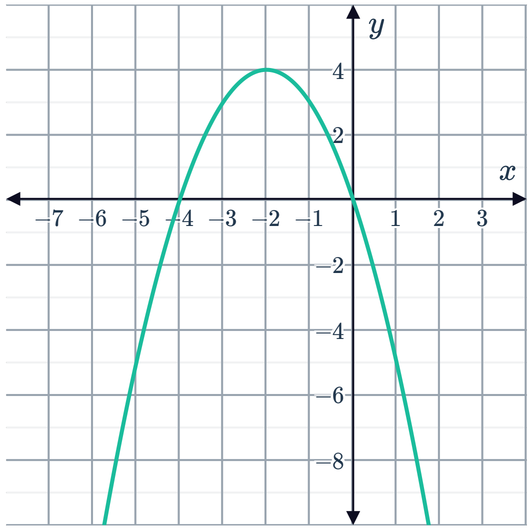
c

i $x = 0, -4$

ii $y = 0$

iii $(-2, 4)$

iv



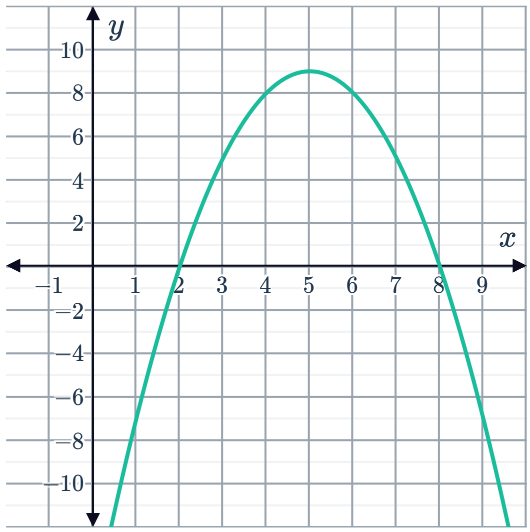
d

i $x = 8, 2$

ii $y = -16$

iii $(5, 9)$

iv



13

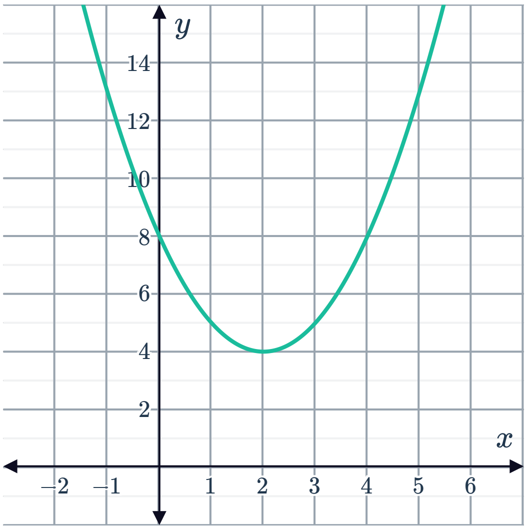
a

i $x = 2$

ii $(2, 4)$

iii $(0, 8)$

iv



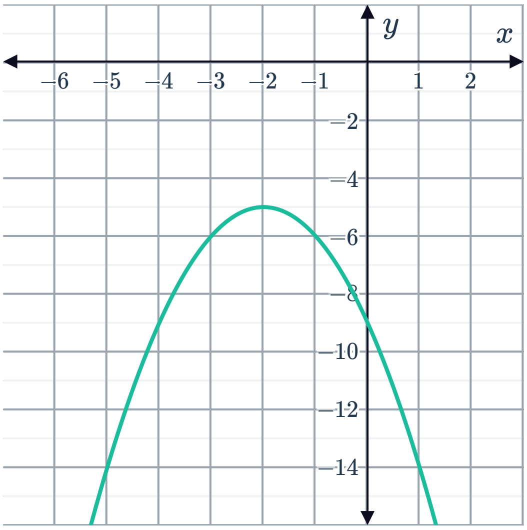
b

i $x = -2$

ii $(-2, -5)$

iii $(0, -9)$

iv



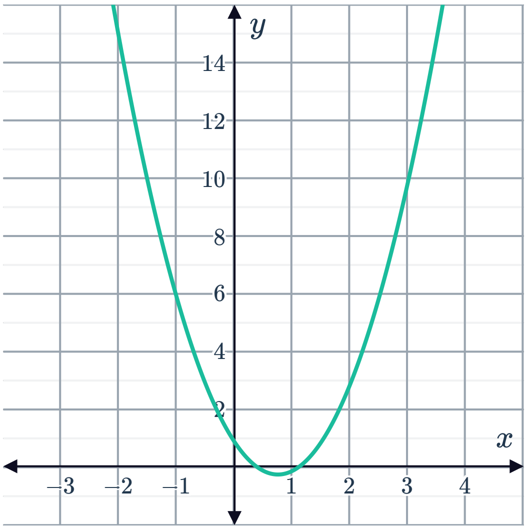
c

i $x = \frac{3}{4}$

ii $\left(\frac{3}{4}, -\frac{1}{4}\right)$

iii $\left(0, \frac{7}{8}\right)$

iv

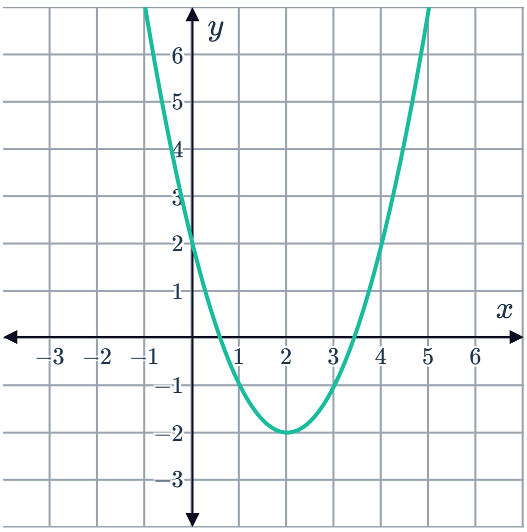


14

a

i $(2, -2)$

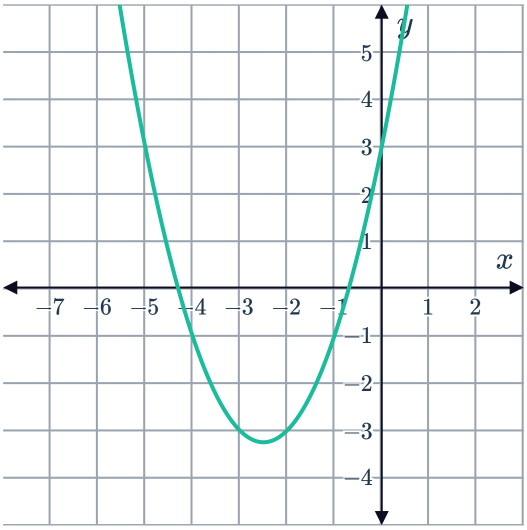
ii



b

i $(-2.5, -3.25)$

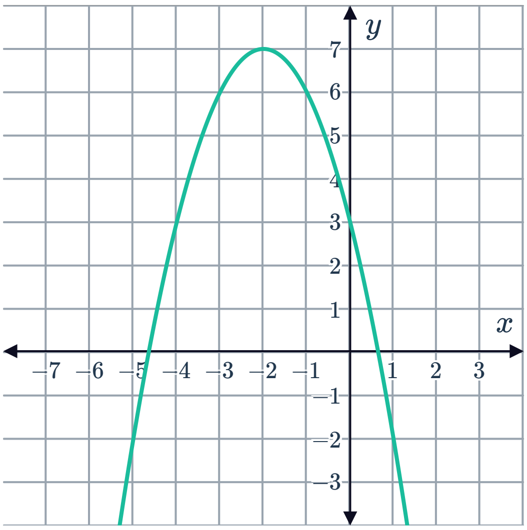
ii



c

i $(-2, 7)$

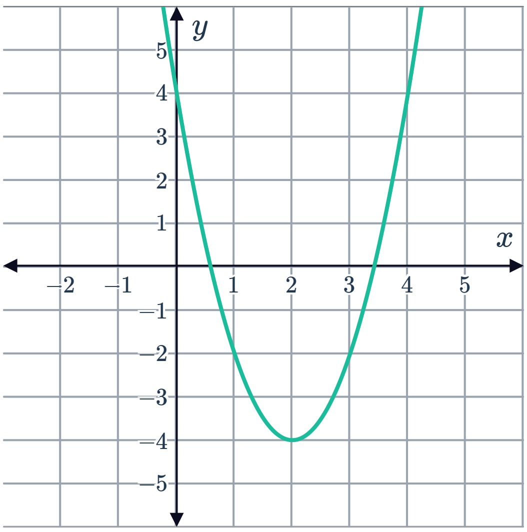
ii



d

i $(2, -4)$

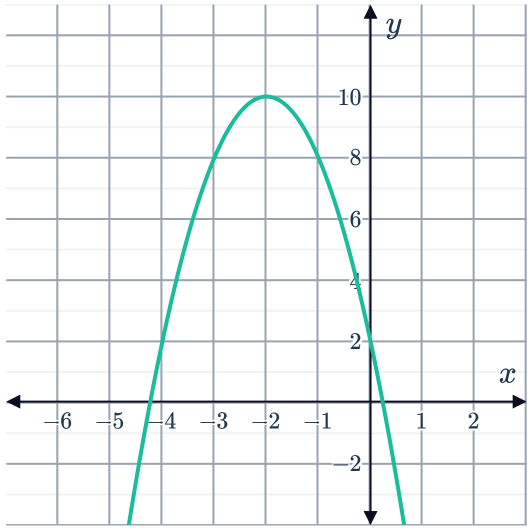
ii



e

i $(-2, 10)$

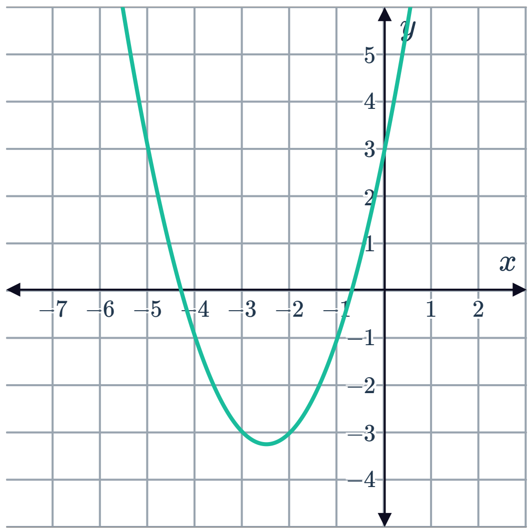
ii



f

i $(-2.5, 4.5)$

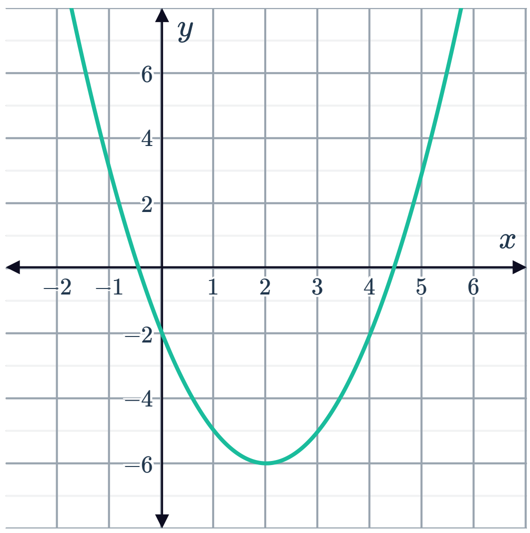
ii



15

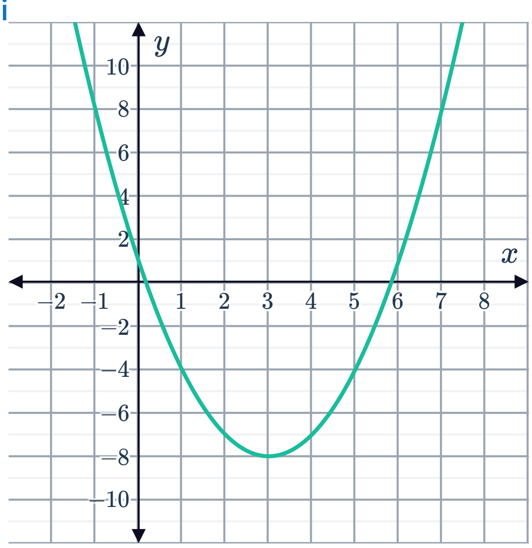
a $y = (x - 2)^2 - 6$

b

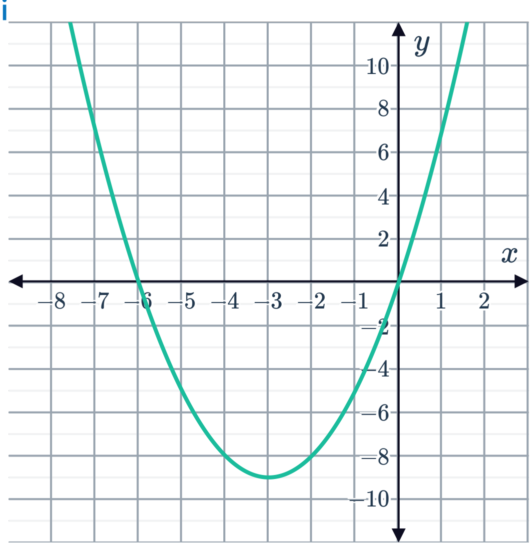


16

a i $(x-3)^2-8$ ii $(3,-8)$ iii



b i $(x+3)^2-9$ ii $(-3,-9)$ iii



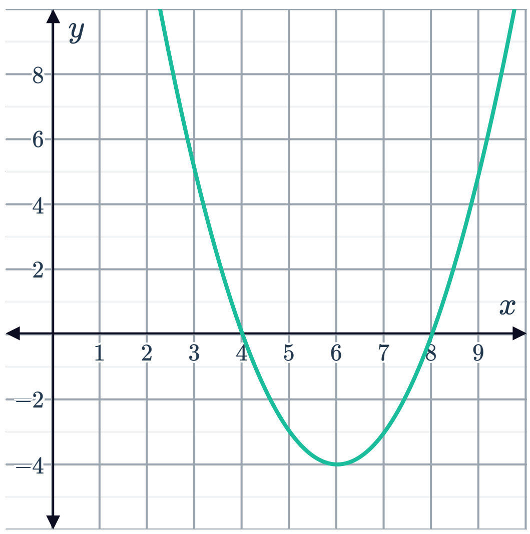
17

a $x=8,4$

b $y=(x-6)^2-4$

c $(6,-4)$

d



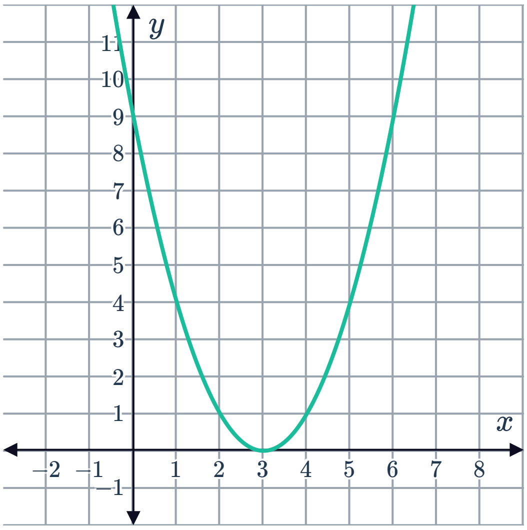
18

a

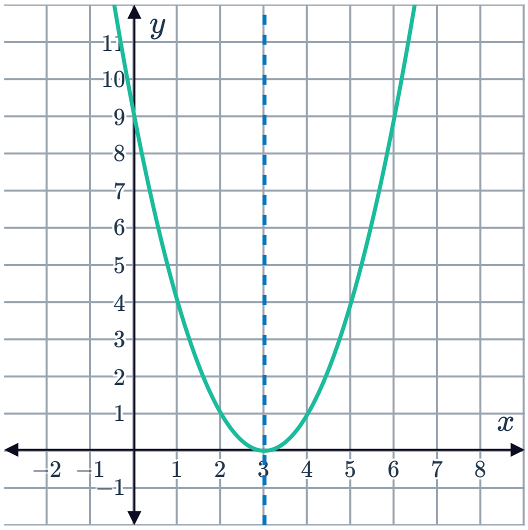
i $(3, 0)$

ii $x = 3$

iii



iv

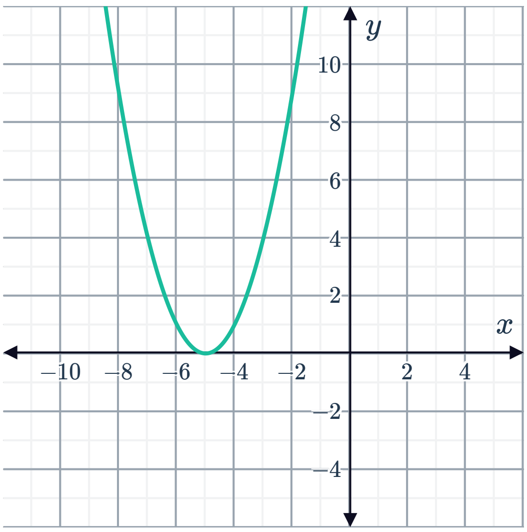


b

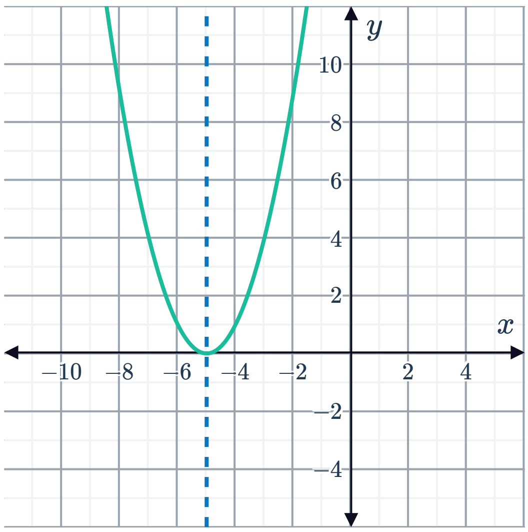
i $(-5, 0)$

ii $x = -5$

iii



iv

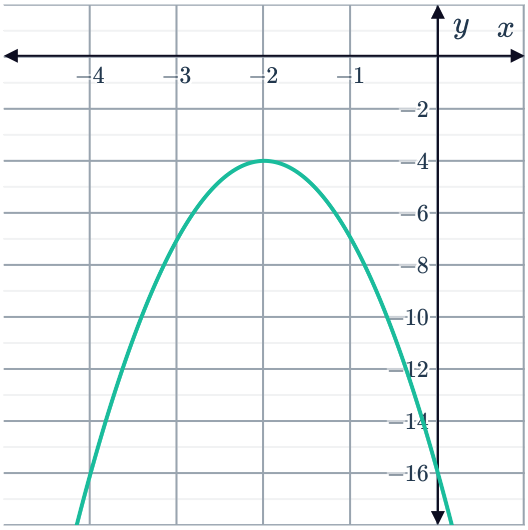


c

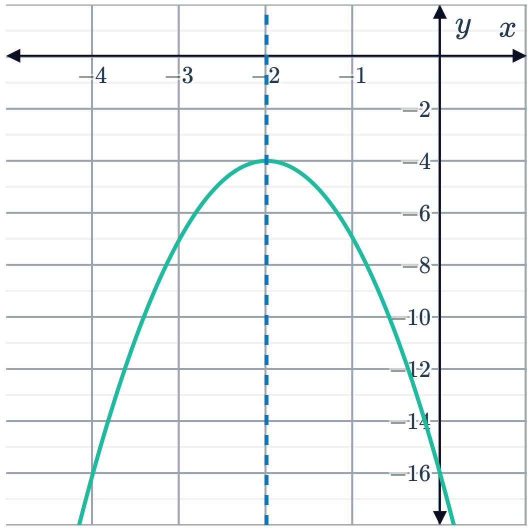
i $(-2, -4)$

ii $x = -2$

iii

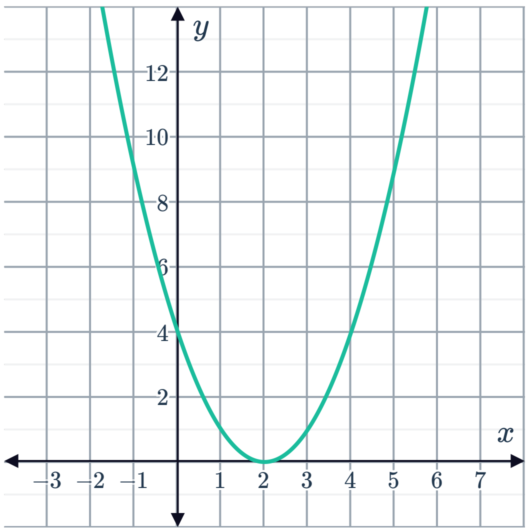


iv

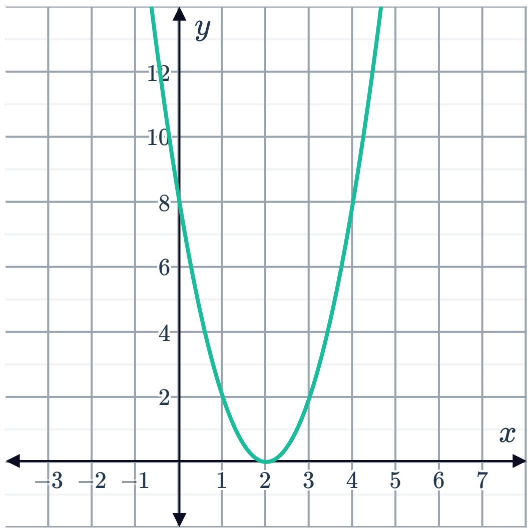


19

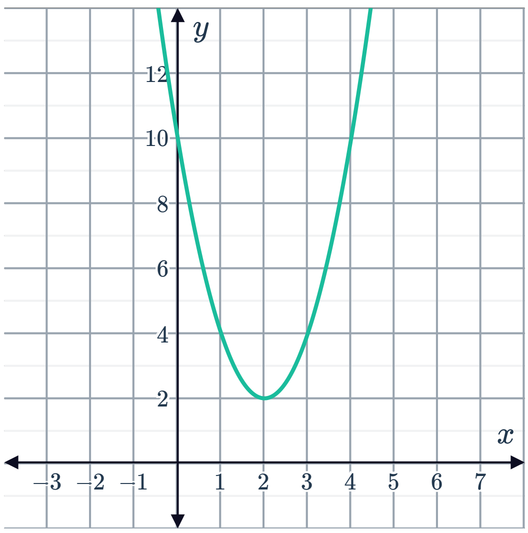
a



b



c



Applications

20

a $s = 12$

b 337

21

a $t = 3$

b 122 units